

Magic Boxes: Curriculum Plan

How the 27 activities fit together — sequencing, timing and equipment

Level: Ages 7–19 • **Time in class:** From a single lesson to a full school year • **Sensors:** BME280, VEML7700, SCD41, MPU-6500

1. What this collection is

Magic Boxes is a sensor platform for school science. A small box containing four sensors streams live measurements to a laptop application, and the application provides two kinds of activity:

- **15 guided exercises** — ordered steps that walk a class through a specific investigation, with theory, instructions and a place to record observations. These are lesson-shaped: start, middle, end.
- **12 apps** — open instruments and physics games. Four are games where the box is a control stick; eight are dashboards for open investigation and long-running projects.

Every activity has **two documents**: a **teacher guide**, covering curriculum links, a time plan, the materials list, the science background, the lesson flow, analysis and discussion questions, common pitfalls, safety notes, adaptations and resources; and a separate **pupil sheet**, two pages long, printed double-sided, one per child. Both exist **in Romanian and in English**, and the Romanian version is the one written first.

The one thing that matters most

Whatever activity you choose, ask pupils to **write down a prediction before any measurement begins**, with a reason. The pedagogical value of these boxes lies almost entirely in the gap between what pupils expect and what the sensor reports. A measurement that merely confirms an unstated expectation teaches very little; a measurement that contradicts a written prediction is remembered for years.

2. The three levels

Activities are labelled by level inside the app. The levels describe *cognitive demand* rather than strict ages, and a capable younger class can attempt the next level up with support.

Level	Ages	What is expected of pupils
Junior	7–10	Read a number, compare two numbers, describe a change in words. Qualitative reasoning: bigger, smaller, faster, warmer.
Explorer	11–14	Read a graph, identify a trend, control a variable, take a systematic series of readings, express a simple relationship.
Scientist	15–19	Calculate a rate or gradient, transform data (logarithms), fit a line, evaluate assumptions and sources of error, connect to formal theory.

3. The full catalogue

3.1 Guided exercises

Activity	Level	Time	Core idea
Light is Everywhere	Junior	15 min	Light varies enormously between places; measurement beats intuition.
The Magic Thermometer	Junior	15 min	Temperature differs from place to place and changes over time.
The Air We Breathe	Junior	15 min	Breathing changes the air; CO ₂ is invisible but measurable.
Flying Straight	Junior	15 min	Orientation and tilt can be measured.
The Box in Balance	Junior	15 min	Gravity gives a reference direction.
Indoor Pollution	Explorer	20 min	CO ₂ indicates ventilation quality.
The Seismometer	Explorer	20 min	Vibrations can be recorded and read.
Pressure and Altitude	Explorer	20 min	Pressure falls predictably with height.
Urban Heat Islands	Explorer	25 min	Built surfaces are warmer than green ones.
Air Convection	Explorer	20 min	Warm air rises; gradients form.
The Water Cycle	Explorer	20 min	Evaporation and condensation, humidity and dew point.
CO ₂ and Photosynthesis	Scientist	30 min	Net versus gross rates; inference from two measurements.
Solar Panels	Scientist	30 min	Incidence angle governs collected energy.
Physics of the Atmosphere	Scientist	30 min	The barometric formula and scale height.
The Beer–Lambert Law	Scientist	30 min	Exponential attenuation; linearising by taking logarithms.

3.2 Apps: games and instruments

Activity	Type	Ages	Core idea
Among the Clouds	Game	all	Tilt control; the accelerometer senses gravity.
Earthquake!	Game	all	Real earthquake safety; P waves precede S waves; the door-frame myth.
Resonance	Game	all	Natural frequency; resonance comes from physics, not from force.
Gyro Target	Game	all	Precision, steadiness, and how a reflex is re-learned.
Measurement book	Note- Instrument	all	Systematic labelled spot readings.
Weather Station	Instrument	all	Live atmospheric monitoring.
Air Quality	Instrument	all	Classroom CO ₂ , several boxes at once.
Solar Energy	Instrument	11–19	Light as an energy resource.
Seismology	Instrument	11–19	A working seismograph.
Urban Heat Islands	Instrument	11–19	Multi-location temperature survey.
Greenhouse Effect	Instrument	11–19	CO ₂ and light together.
Spectroscopy	Instrument	15–19	Transmission and absorption of light.

4. How much time does it take?

Three realistic scales of adoption. Choose one and plan equipment accordingly.

Scale	Total time	What it looks like
Taster	1 lesson (50 min)	One game to hook the class (10 min), one junior exercise done properly with predictions (25 min), discussion (15 min). Requires no preparation beyond charging a box.
Unit	6 lessons over 2–3 weeks	A themed sequence: for example air quality — The Air We Breathe, Indoor Pollution, a week of Air Quality monitoring, then CO ₂ and Photosynthesis, ending with pupil presentations.
Full programme	20–30 lessons across a school year	The progression in the next section, one activity every week or fortnight, with two extended projects and a science-fair style showcase.

A single guided exercise occupies 15–30 minutes of app time, but plan a **full 50-minute lesson** around it: predictions, setup, measurement, export and discussion. The stated durations are the measurement phase only.

5. A year-long progression

The sequence below builds skills deliberately. Each stage ends with an open-ended project in which pupils choose their own question — that is where the earlier structured work pays off.

The three stages each correspond to roughly a third of the year and can be laid over the school's own learning modules, however these are divided.

5.1 Stage 1: measuring and comparing

1. **Among the Clouds** (game) — introduce the hardware playfully. Pupils discover that the box knows how it is being held.
2. **Light is Everywhere** — the first real measurement. Skill: reading a value and comparing places.
3. **The Magic Thermometer** — skill: repeated measurement, recognising that values change over time.
4. **The Air We Breathe** — skill: measuring something invisible. First encounter with a sensor that reveals what the senses cannot.
5. **Measurement Notebook** (app) — skill: systematic labelled recording. Pupils design their own comparison of five places.
6. **Project 1 (2 lessons)**: a survey of the school. Groups choose a quantity, measure it in eight locations, and present a ranked chart.

5.2 Stage 2: relationships and variables

1. **Indoor Pollution** — skill: interpreting a trend over time and connecting it to a cause (occupancy, ventilation).
2. **Air Convection** — skill: controlling a variable, measuring at different heights.
3. **Pressure and Altitude** — skill: a systematic series and a near-linear relationship. Best done in a stairwell.
4. **The Water Cycle** — skill: two quantities changing together.
5. **Urban Heat Islands** (the exercise, then the app) — skill: field measurement and fair comparison between sites.

6. **Earthquake!** and **Resonance** (games) — a change of pace, with a serious debrief on earthquake safety and on natural frequency.
7. **Project 2 (3 lessons)**: a week-long monitoring campaign. Groups run Air Quality or Weather Station daily, export each day, and compare days. This is where day-to-day file comparison becomes real.

5.3 Stage 3: quantitative science

1. **The Seismometer** and the **Seismology** app — skill: event detection in a noisy signal.
2. **Solar Panels** — skill: an angle series, and a cosine relationship.
3. **CO₂ and Photosynthesis** — skill: calculating a rate, and inferring an unmeasured quantity from two measurements.
4. **Physics of the Atmosphere** — skill: applying a formula to real data, and evaluating how well it fits.
5. **The Beer–Lambert Law** — skill: transforming data with logarithms to obtain a straight line, then fitting a gradient.
6. **Project 3 (3 lessons)**: an independent investigation of the pupils’ own design, written up as a formal report and presented.

6. Themed routes

If you cannot devote a year, these shorter routes each tell a complete story.

Route	Lessons	Activities in order
Air and health	4	The Air We Breathe → Indoor Pollution → Air Quality (week-long) → CO ₂ and Photosynthesis.
Climate and energy	5	Light is Everywhere → Solar Energy → Solar Panels → Greenhouse Effect → Urban Heat Islands.
Earth in motion	4	Among the Clouds → The Box in Balance → The Seismometer → Earthquake! and Resonance.
Weather and atmosphere	5	The Magic Thermometer → Weather Station → The Water Cycle → Pressure and Altitude → Physics of the Atmosphere.
Light and matter	4	Light is Everywhere → Solar Energy → Spectroscopy → The Beer–Lambert Law.

7. The three-strand assessment rubric

Every guide refers to this rubric and none of them repeats it: it is defined here, once, so that twenty-seven versions of it cannot drift apart.

Three strands are worth tracking over a year, because the activities develop them cumulatively:

- **Measurement practice** — does the pupil take repeated readings, record units and conditions, and recognise when a value is not yet stable?
- **Data interpretation** — can the pupil describe a trend, read a gradient, distinguish a real change from noise, and compare two datasets fairly?
- **Scientific reasoning** — does the pupil predict before measuring, explain a discrepancy, identify assumptions, and propose a test rather than an opinion?

Strand	Emerging	Secure	Confident
Measurement practice	Reads a value off the screen and writes it down. Needs reminding about units and repeats.	Repeats the measurement, records the units and conditions, waits for the value to settle, and changes one variable at a time.	Designs the series unprompted: how many repeats are needed, what is held constant and why, and recognises a compromised reading before using it.
Data interpretation	Says which value is larger and in which direction it changed.	Describes the trend of a graph, links it to the likely cause, and compares two series fairly.	Quantifies: gradient, rate, order of magnitude. Separates noise from signal and says how confident the conclusion is.
Scientific reasoning	Makes a prediction when asked and notices whether it held.	Predicts with a reason, explains the mismatch between prediction and result, and proposes a check.	Names the assumptions behind the method, proposes a test that could disprove their own explanation, and separates the instrument's limits from the theory's.

Beware a clash of words. The three columns above are *levels of attainment* and have nothing to do with the *Junior*, *Explorer* and *Scientist* levels of the activities. A middle-school pupil can be *confident* on a Junior activity; a sixth-former can be *emerging* on their first open investigation. The rubric describes the pupil, not the activity.

How to use it. Not every lesson. Choose **one strand** per activity — each activity's guide says which one shows up most clearly there — and follow that. Over a year, every pupil will have several observations on each strand, which is more than any test gives.

The best form of portfolio is still the pupil's own folder of exported files with annotated graphs, built up over the year. It shows progression more honestly than any mark.

8. Equipment planning

8.1 How many boxes?

Boxes	Class of 25	What is possible
1	Whole class	Demonstration on a projector; games with pupils taking turns; a single running monitor for a week-long project.
3–4	Groups of 6–8	Rotation through stations. Workable, but the groups are large for the more fiddly setups.
6–8	Groups of 3–4	The comfortable minimum for practical work where everyone handles equipment.

Two activities — Air Quality and Urban Heat Islands — deliberately support several boxes at once and become considerably richer with three or more.

8.2 Consumables and general equipment

The per-activity guides list exactly what each one needs. Across the whole collection, the recurring items worth having in a box on the shelf are: a desk lamp; sheets of paper and card; a torch;

coloured filters or gel sheets; food colouring or ink; transparent containers with lids; a jug and a kettle; ice; cling film; a cloth or blanket for creating darkness; a tape measure and a ruler; a protractor; clipboards; sticky labels; and two or three robust houseplants.

8.3 Before the first lesson

1. Install the application and connect one box, either by USB cable or over WiFi, and confirm live values appear.
2. Run one activity yourself, end to end, including the export. Twenty minutes spent here prevents most classroom problems.
3. Check where the data is saved. Sessions are written automatically to the `logs` folder next to the application as spreadsheet-ready CSV files, one column per sensor, and the Export button in each app saves a formatted Excel workbook with a date-stamped name. Decide *now* where pupils will keep their files, because day-to-day comparison depends on it.
4. **Print the pupil sheet** for the activity you have chosen — it is a separate document from the guide, two pages long, meant to be printed double-sided, one sheet per child.

Programme-level pitfalls

Treating the sensor as an oracle. Every reading has a warm-up period, a noise level and a failure mode. Teach pupils to ask whether a number is plausible before they trust it. A sensor that disagrees with expectation is sometimes making a discovery and sometimes simply wrong; distinguishing the two is the scientific skill.

Playing the games without debriefing them. Twenty minutes of *Earthquake!* without the discussion afterwards teaches pupils that earthquakes are entertaining, and may leave the door-frame myth intact. **The debrief is not optional.**

Skipping the prediction. It takes 90 seconds and it is the single highest-value habit in the whole programme.

Losing the data. Exports are per session. Establish a file-naming and storage convention in the first lesson, before there is anything to lose.

Over-scripting. The exercises are structured on purpose, but the lasting learning tends to come from the open projects. Protect that time.

General notes

Individual guides carry activity-specific safety information; read the relevant one before teaching. Across the collection, the recurring considerations are trailing cables and power leads; lamps that become hot; hot water and kettles; electrical equipment near water; supervision ratios and route planning for outdoor surveys; and secure handling of the boxes themselves during motion games, where a dropped unit is the most likely damage. Follow your own institution's risk-assessment procedures, which take precedence over anything written here.

9. Where to find things

Item	Location
Teacher guides	<code>docs/ro/ghiduri_profesor</code> in Romanian and <code>docs/en/teacher_guides</code> in English, one PDF per activity.
Pupil sheets	<code>docs/ro/fise_elev</code> and <code>docs/en/student_sheets</code> , two pages each, to be printed double-sided.
Technical manual	<code>docs/ro/manual_tehnic</code> and <code>docs/en/technical_manual</code> — sensors, protocol, file formats.
Automatically saved data	The <code>logs</code> folder next to the application, as CSV files named by box and date; high-rate motion data is written to a separate file ending in <code>_imu</code> .
Exported data	Wherever you choose in the Export dialogue; the suggested name carries the box identifier and the date.
Version	Shown on the application's home screen and in Settings. These documents describe version 5.2.